

Saint Petersburg State
University Graduate School of Management

End-to-end Sample ML Project course

Report 1: Exploratory Data Analysis (EDA)

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Dataset Overview

The dataset was downloaded from Google Drive using curl command and contains vehicle auction sales records from the US market.

Property	Value
Total rows	50,000
Total columns	15
Target variable	sellingprice
Task type	Regression
Source	Kaggle - Vehicle Sales Data

Features Description

Feature	Type	Description
year	Numerical	Year of vehicle manufacture
make	Categorical	Vehicle brand (Ford, Chevrolet, etc.)
model	Categorical	Vehicle model name
trim	Categorical	Trim level / configuration
body	Categorical	Body type (Sedan, SUV, etc.)
transmission	Categorical	Transmission type (automatic/manual)
vin	Text	Unique vehicle identifier - not used
state	Categorical	US state of sale
condition	Numerical	Vehicle condition score (1-49)
odometer	Numerical	Mileage in miles
color	Categorical	Exterior color
interior	Categorical	Interior color
seller	Categorical	Seller name
sellingprice	Numerical	TARGET: sale price in USD
saledate	Text	Date of sale - not used

Missing Values Analysis

The following missing values were identified in the dataset:

Feature	Missing Count	% Missing
transmission	5,857	11.7%

body	1,194	2.4%
trim	976	2.0%
model	939	1.9%
make	933	1.9%
condition	1,046	2.1%
color	72	0.1%
interior	72	0.1%
odometer	9	0.0%
sellingprice	4	0.0%

Imputation Strategy

- Categorical columns (make, model, trim, body, transmission, color, interior): filled with mode (most frequent value)
- Numerical columns (condition, odometer): filled with median
- Rows with missing sellingprice (target): dropped

EDA Visualizations

Distribution of Selling Price

The selling price distribution is right-skewed, with most vehicles priced between \$5,000 and \$25,000. The median price is \$12,200. There are some high-value outliers above \$60,000.

Top 10 Car Makes

Ford is the most frequently sold brand (~6,400 vehicles), followed by Chevrolet (~3,800) and Nissan (~3,400). Toyota, Dodge, and Honda round out the top 6.

Average Selling Price by Year

There is a clear positive trend: newer vehicles sell for significantly more. Cars from 2015 average ~\$25,000, while pre-2000 vehicles average under \$5,000.

Odometer vs Selling Price

A clear negative relationship: as mileage increases, selling price decreases. High-priced vehicles tend to have low odometer readings. The relationship is non-linear.

Correlation Heatmap

Key correlations with sellingprice:

- year: +0.58 (strong positive - newer = more expensive)
- odometer: -0.58 (strong negative - more miles = cheaper)
- condition: +0.31 (moderate positive - better condition = higher price)

Feature Importance Conclusions

Most important features for sellingprice prediction:

- year - correlation 0.58 with price. Newer cars are significantly more expensive.
- odometer - correlation -0.58 with price. More mileage = lower price.
- condition - correlation 0.31. Better condition = higher price.
- make / model - brand strongly affects price (Ford vs BMW)

Features that are hard to process:

- vin - unique identifier per car, not useful for prediction - dropped
- saledate - text format, needs parsing - dropped
- seller - too many unique values, hard to encode effectively
- trim, model - hundreds of unique values, need careful encoding